5 Assembly and Dismantling R Food

DANGER

Risk of injury

Improper handling of the packaged door can lead to injury.

⇨ Place palette on an even surface.

⇨ Do not remove palette before starting assembly.

5.1 Inspection of the delivery

The door is largely pre-assembled on delivery. The side frames, top roll cover, motor cover (if supplied), fixing materials and control system are secured to a transport frame (wooden palette).

- Check door for damage incurred during transit
- Check that the delivery is complete by comparing it with the order documentation.

5.2 Preparation for assembly

DANGER

Risk of injury

Incorrect assembly of the door can lead to injury.

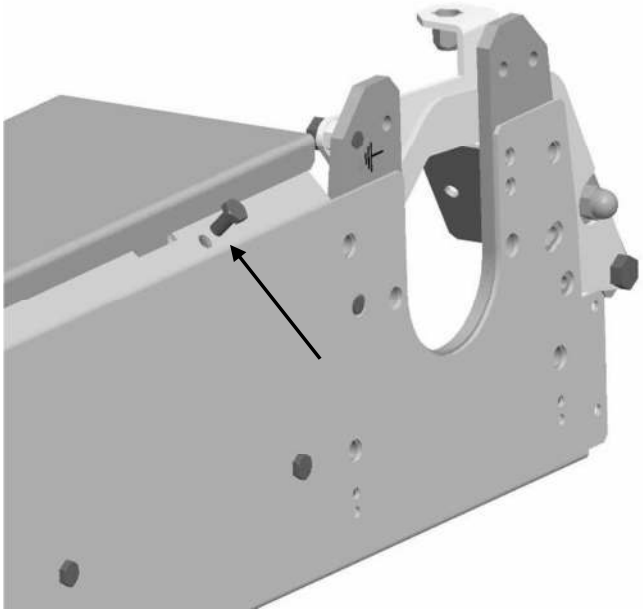
Isopanel walls, chequer bricks and light partition walls must be checked for their suitability for assembly.

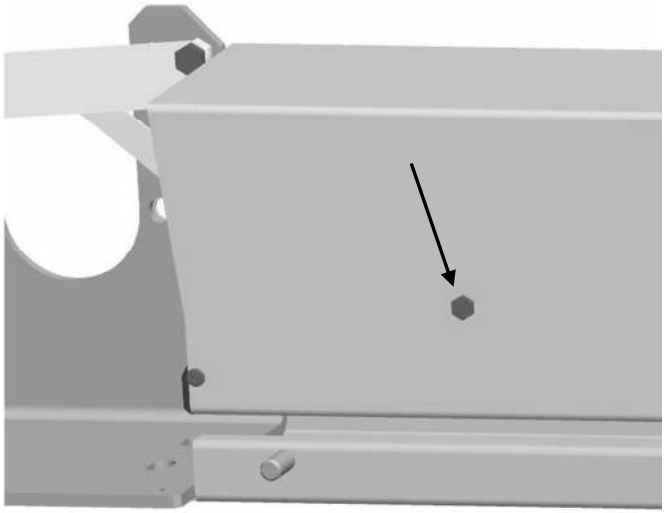
⇨ The door may only be assembled by trained staff.

⇨ The door may only be assembled by the manufacturer or an installation firm approved by the manufacturer.

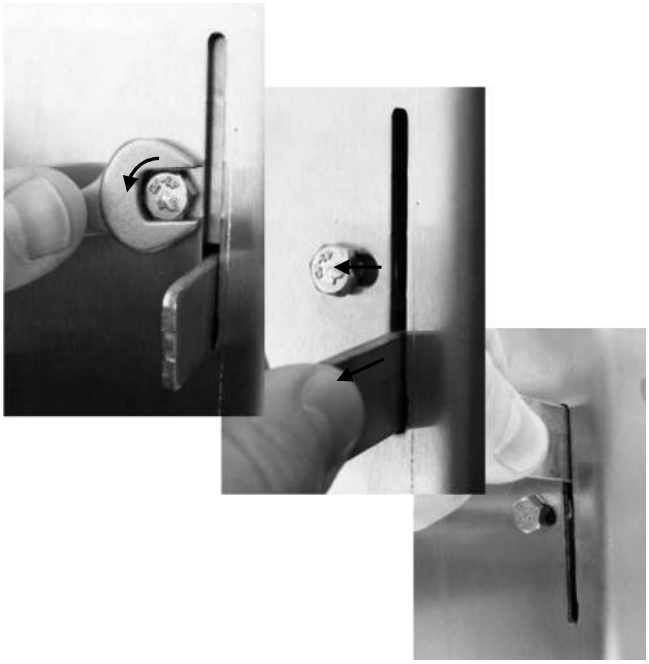
Make the following preparations before assembling the door:

- Inspect the installation site for the necessary fixing points and assembly requirements: assemble a substructure or other suitable fixing options if necessary. Screw the upper fixing points to a steel substructure or concrete element. The fixing plan can be found in the user manual.
- Measure clearance dimensions (width and height) of the door opening and compare with the order.
- Cordon off the installation site against people and vehicle movements.
- Remove shrink-wrap from the packaged door.

	
1. The door shipment must be checked for transport damages and completeness – compare with delivery papers / order papers.	**2.** The drive and fixing material are delivered in a cardboard box.
	
3. Disassemble the covers of the side columns.	**4.** To remove the cover, remove the single retaining screw and the cap nut on each side frame. Save these retaining screws, cap nuts and retainer brackets for later use.



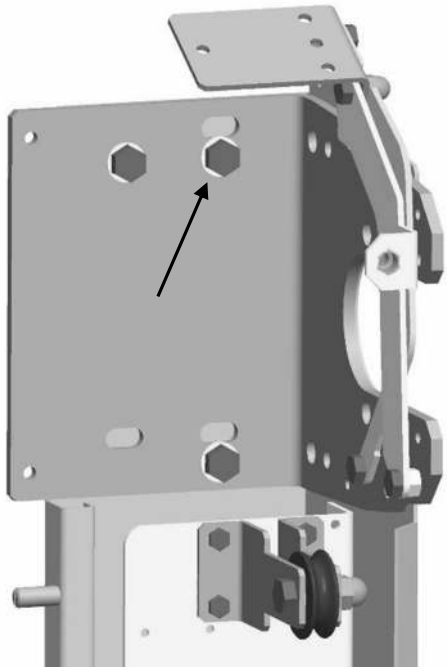
5. Side frame without easy-opening option: To remove the cover, remove the screws on the inside. Lift the cover by approximately 45° and pull it fre e from the side frame profile.



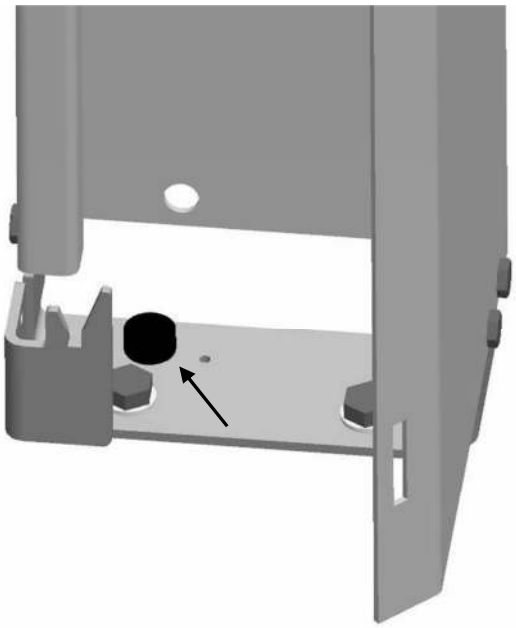
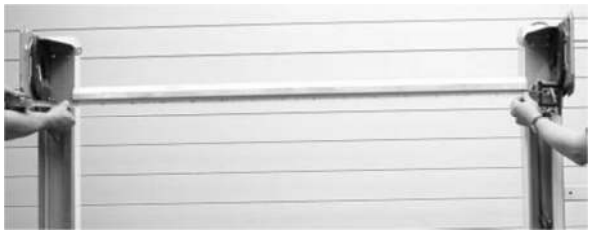
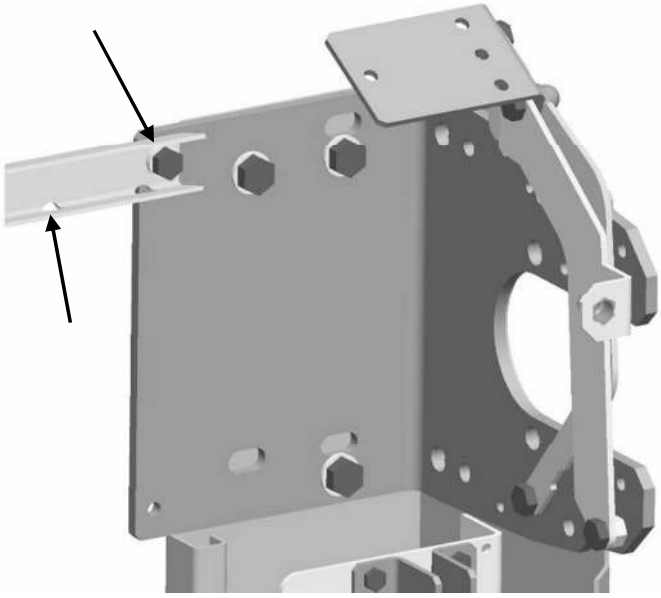
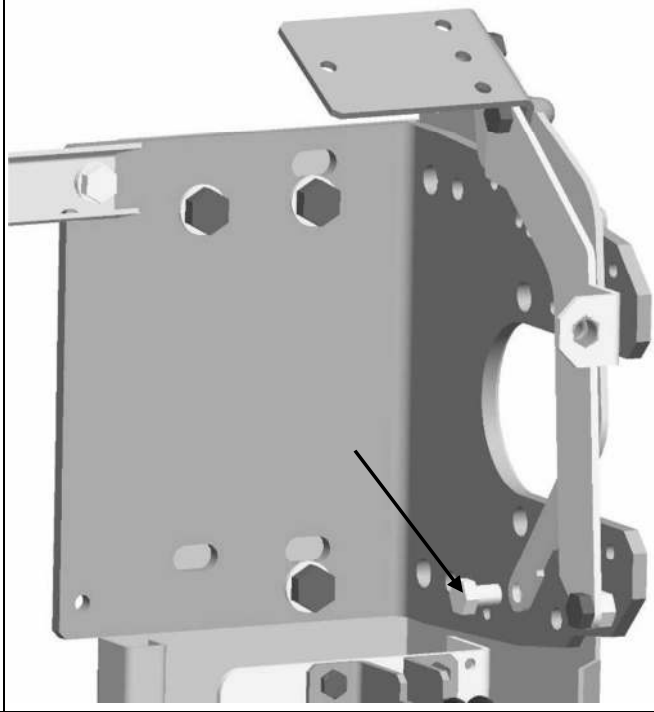
6. Side frame with easy-opening option: To remove the cover, release the screw and move it to the outer side. Pull out the lever and slide it up to the top plate (head plate). Lift the cover by approximately 45° and pull it free from the side frame profile.

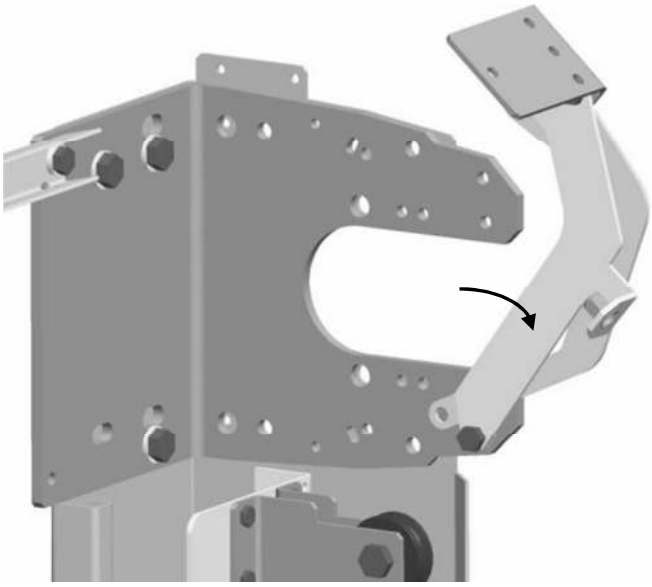


7. If the floor slopes, fit the side frame at the high side first. Position the first side frame and secure it against falling over (e.g. with screw clamps). Use a spirit level or similar tool to align the side frame exactly vertical.

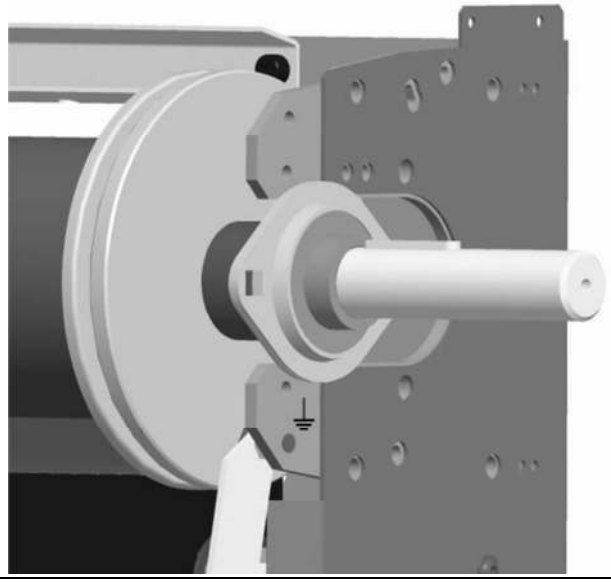


8. Fix the top plates (head plates) to the wall with three screws (use washers). (These screws and washers are not included with the door).

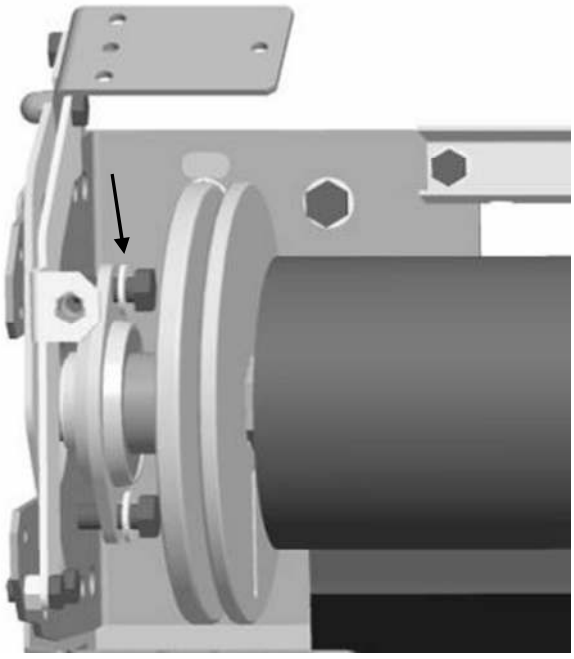
	
9. Fix each bottom plate with two screws (not included). The rubber bumper marked by the arrow is supplied only with doors which have a pre-running photocell.	**10.** Use the cable channel (= exact clearance width) to align the second side frame. First check the length of the cable channel. Maximum height difference between side frames: 3 mm. If necessary, shim the lower side. Width tolerance: ±3 mm.
	
11. Using screws M8 x 12 and washers, fit the cable channel to the side frames at the designated locations. The holes in the cable channel must face downward.	**12. Door with top roll cover, only drive side:** Remove the screw marked by the arrow. Safe this screw for later use.



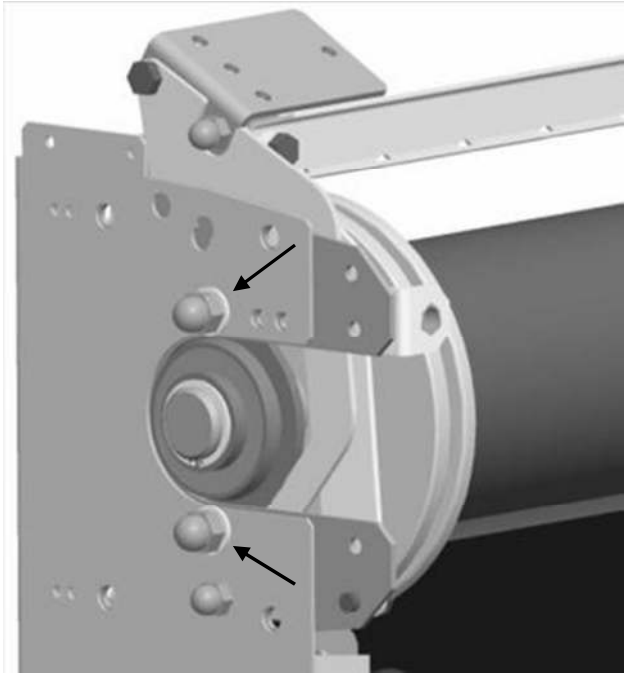
13. Door with top roll cover, only drive side: Pivot the lever downward.



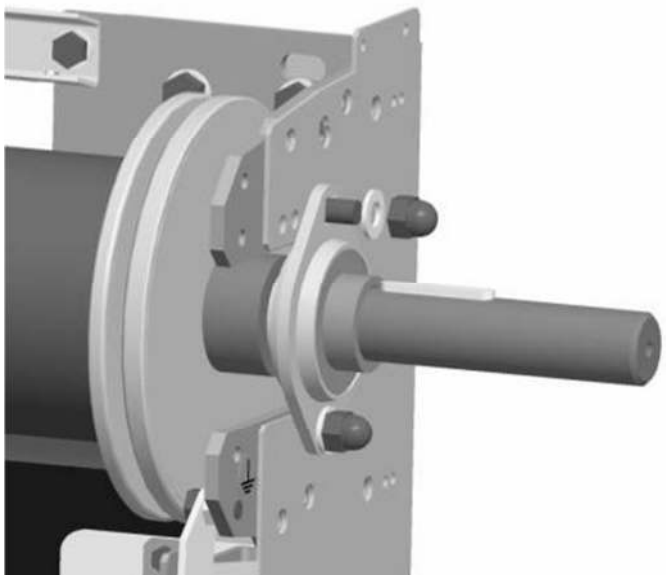
14. Lift-up the top roll with the help of crane or forklift. If the door is small, this can be done manually by hand (2 persons). First position the top roll at the bearing end (short shaft end) and then at the drive end (illustrated) by sliding into the top plate

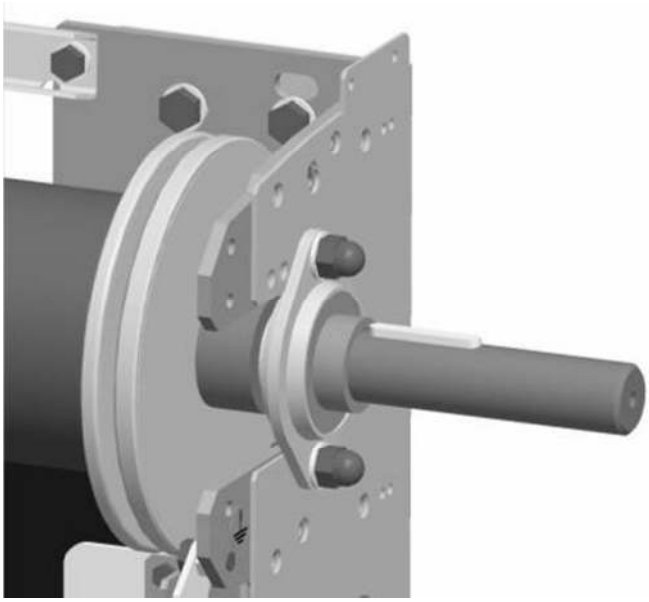


15. Secure the top roll at the bearing end (short shaft end) with two hex head screws M10x25. Use 1 washer per screw. Fit the bearing at the **bearing end** (short shaft end) **from the inside.**



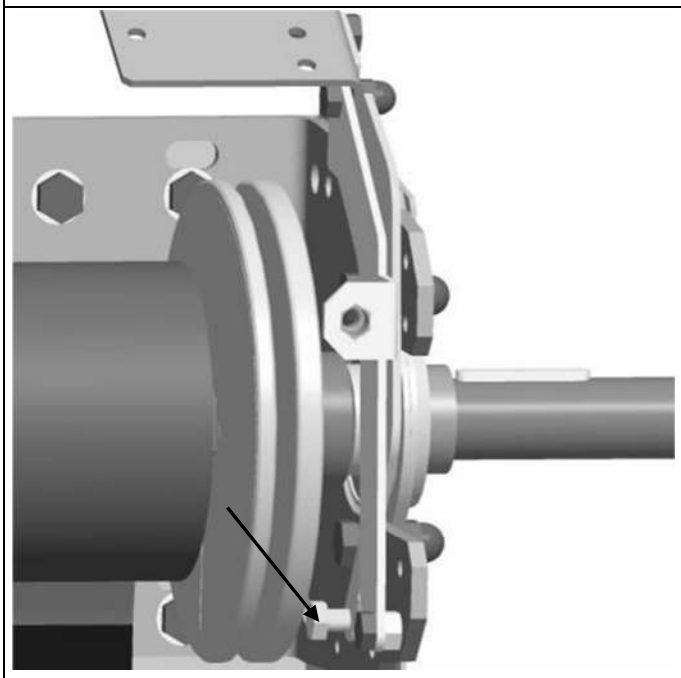
16. Fit washers on the screws and thread on two M10 cap nuts. Tighten the nuts and screws only after the bearing has been fitted on the drive end.

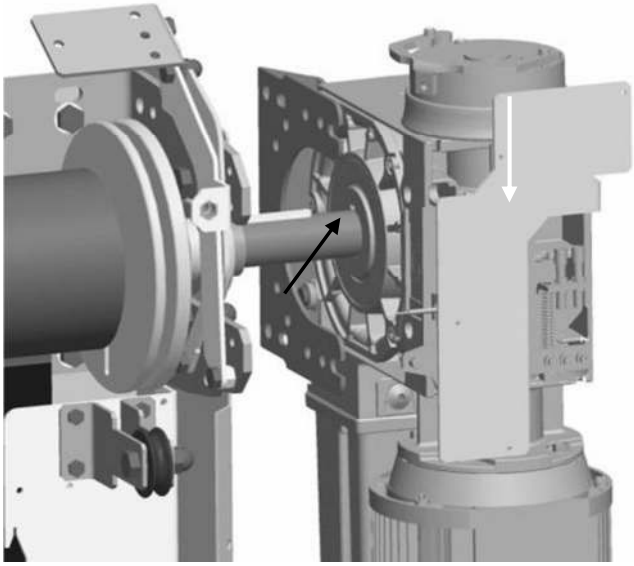




17. The bearing at the **drive end** must be fitted **from the outside.** Secure the bearing with two hex head screws M10x25 and two cap nuts (with washers).

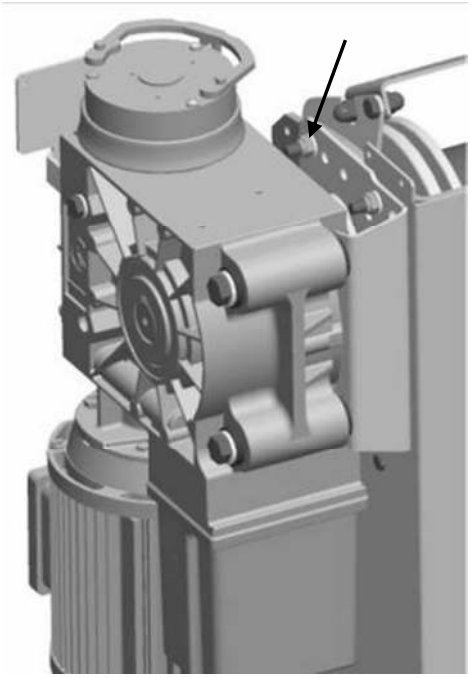
18. Firmly tighten the bearing nuts and screws at the **drive end** and the **bearing end**.



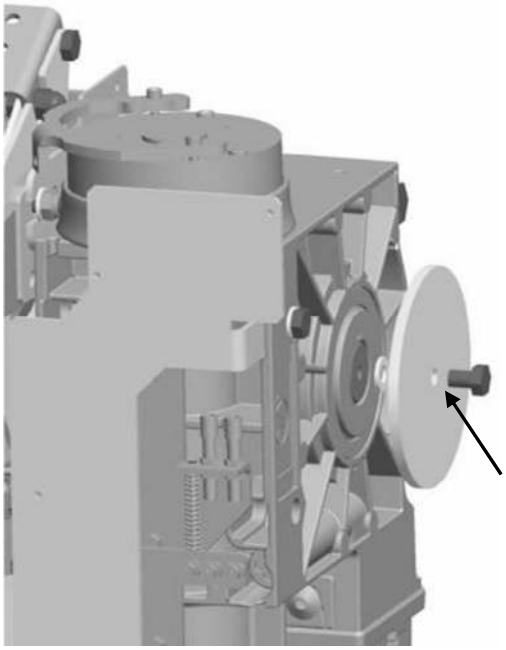


19. Door with top roll cover, only at the drive end: Pivot the lever back and fit the screw marked by the arrow to secure the lever (do not tighten the screw).

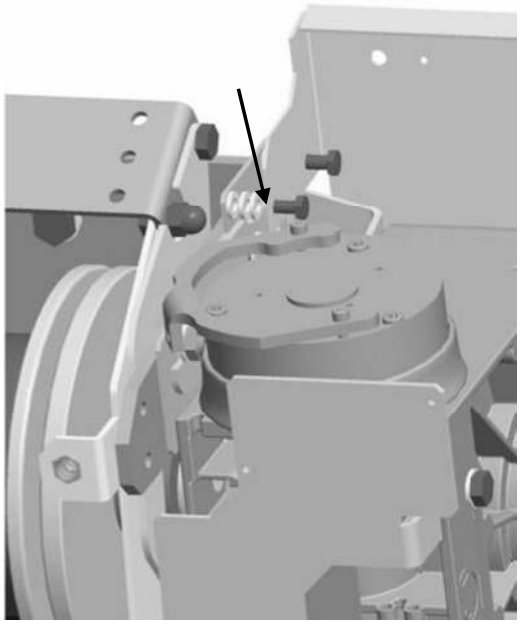
20. Put on the drive; pay attention to the feather key connection.



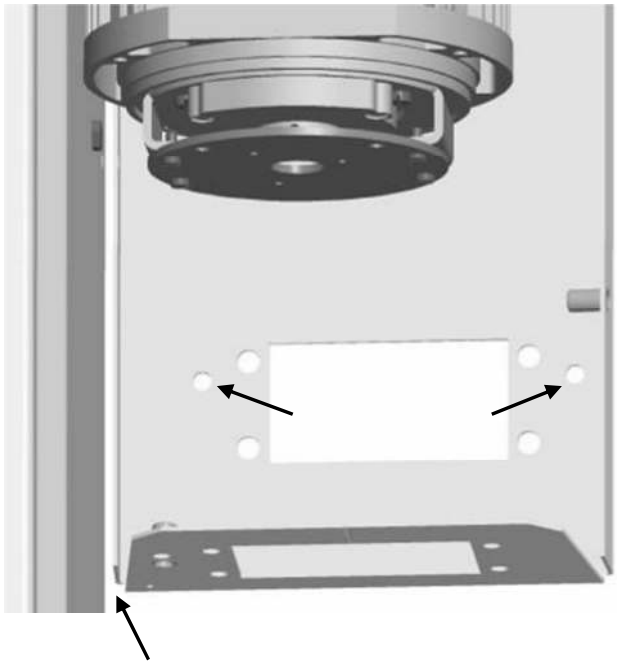
21. Fit the torque arm using M8x25 hex head screws, washers and self-locking M8 nuts. Then tighten the hexagon screw marked in **Fig. 19** against the torque arm.



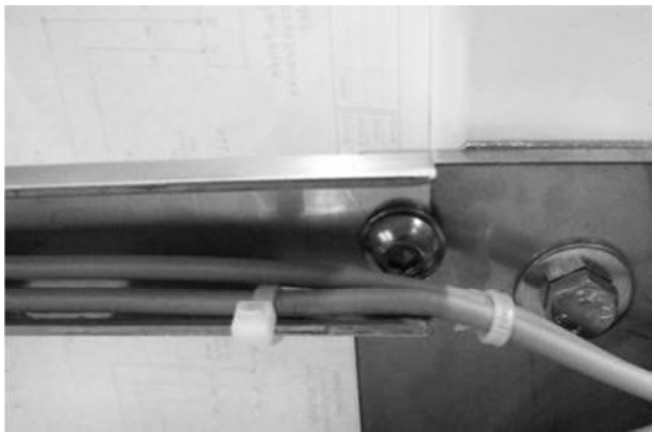
22. Screw the black plastic plate with a hex head screw M8x12 to the shaft on the drive side. Put a washer Ø 8,5 between the shaft and the plastic plate.



23. Door with motor cover: Place three washers Ø 6,5 between the side frame and the motor cover wall bracket (6 pcs. in all). Fasten the motor cover wall bracket to the side frame with two M6x12 hex head screws.



24. Door with motor cover: Fasten the motor cover wall bracket to the wall with four screws (not included with the door). Leave a gap of approximately 4 mm between the side frame and the wall bracket (arrow).



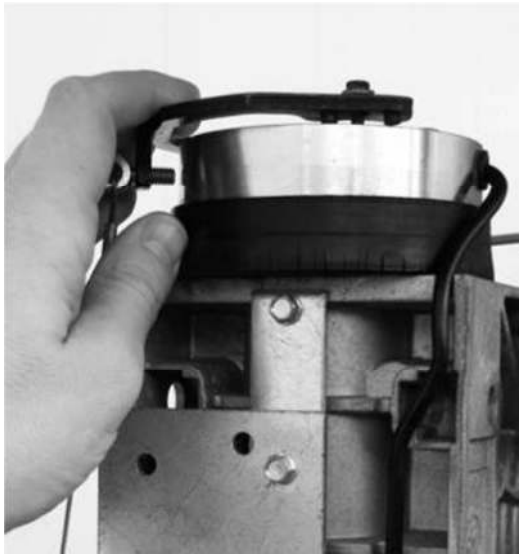
25. Feed the photocell cable(s) through the cable channel from the bearing end to the drive end. Secure the cable(s) to the cable channel with cable ties (see photo).



26. Release the door curtain safety device.



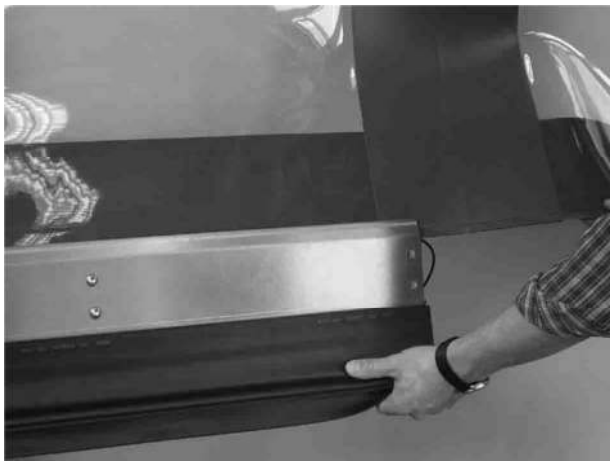
27. Lower the door curtain to working height. You have to release the brake on the drive unit to lower the door curtain. Turn the door curtain by hand. Brake operation: **see Fig. 28.**



28. Operate the brake attached to the gearbox by raising or lowering the black handle.



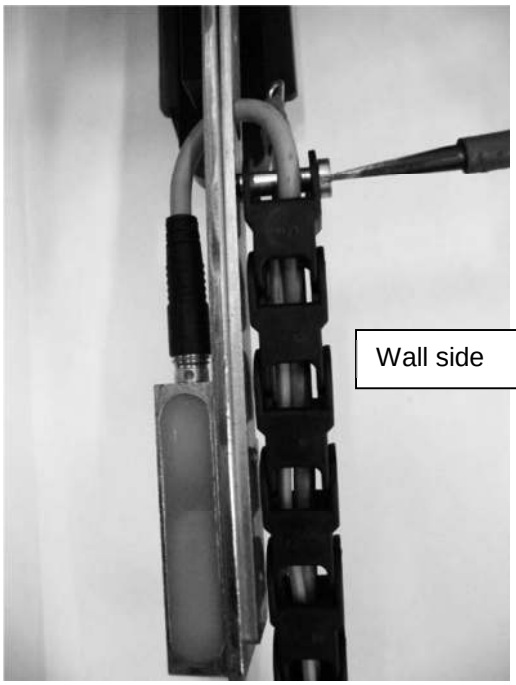
29. Loosen the screws of the bottom profile before sliding the bottom profile onto the door curtain. Do not remove these screws!!



30. Slide the bottom profile onto the bead of the door curtain.



31. Align the bottom profile so it is centered to the door curtain.



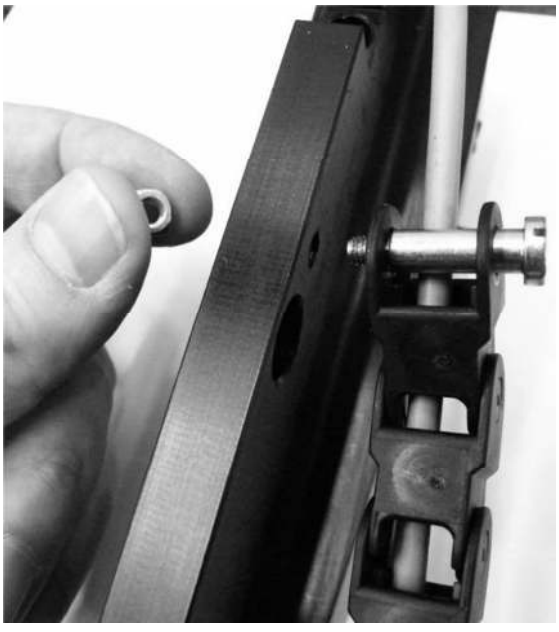
32. Door with pre-running photocell without breakaway: Plug the connector of the cable chain into the pre-running photocell (ensuring it is correctly aligned). Use the flat head screw to attach the cable chain to the slider.



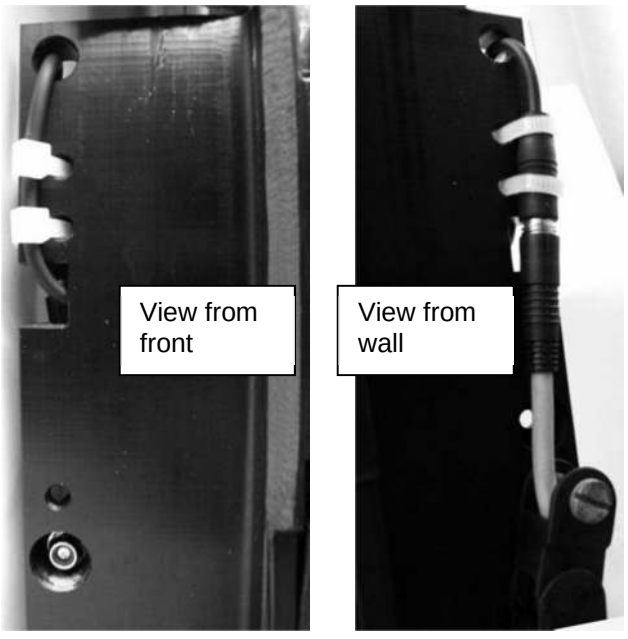
33. Door with pre-running photocell without breakaway: Fasten the cable eye for the equipotential bonding to the slider with a M3x8 screw.



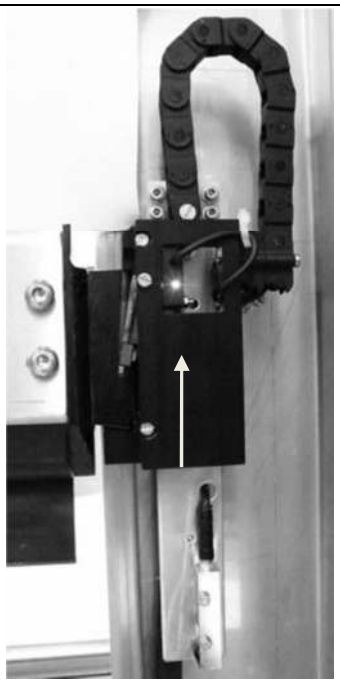
34. Door with pre-running photocell without breakaway: Fasten the slider with four M8x16 hex head screws. Check the free movement of the sliders.



35. Door with safety edge, drive side: Attach the cable chain to the slider with the flat head screw and an M4 nut. Position the cable chain on the wall side. (drive side shown to the right)



36. Door with safety edge, drive side: The cable chain is located behind the slider after installation. (drive side shown to the left). Secure the connectors with two cable ties.

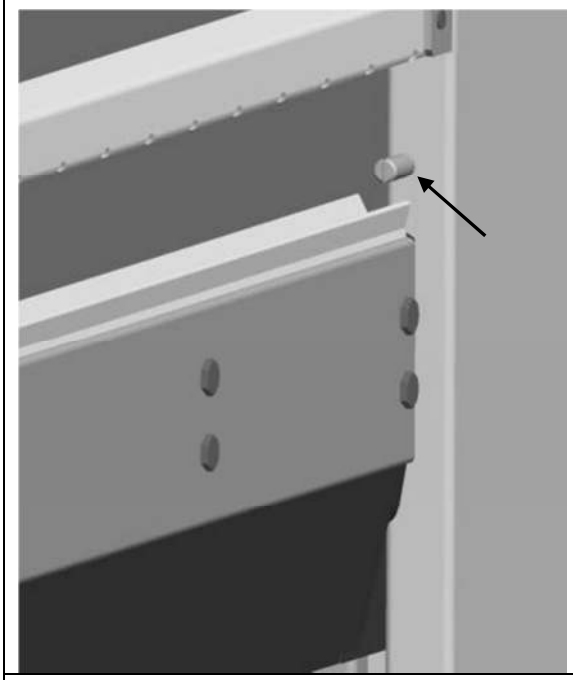


37. Door with pre-running photocell and breakaway: Slide the slider fitted in the side frame from the bottom up to the height of the bottom profile. The detent must engage automatically. Check the free movement of the sliders.



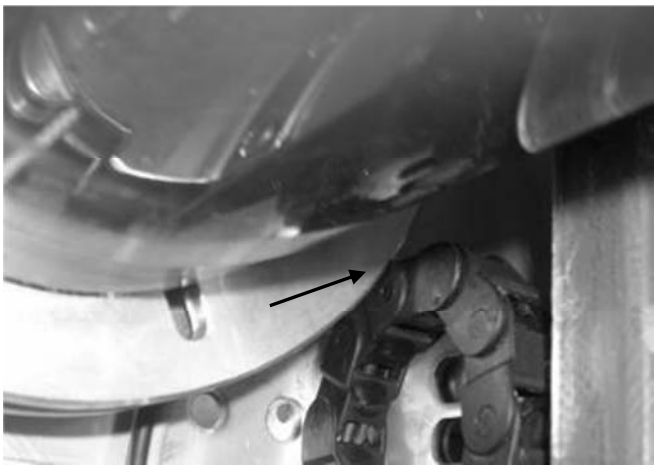
Attention: If the side frame cover is not closed and you open or close the door, the bottom profile can be caught by the stiffeners!

38. Side frame cover not closed

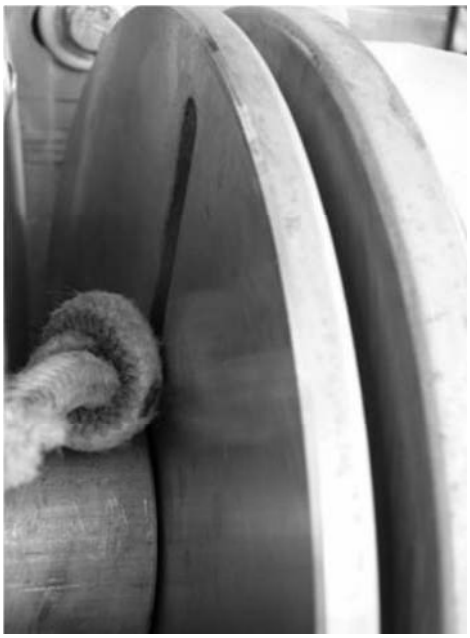


39. Door with balancing system: Open the door exactly up to the daylight height, the flexible lip stays in the opening area. Doors with pre-running photocell have a block at the side frame.

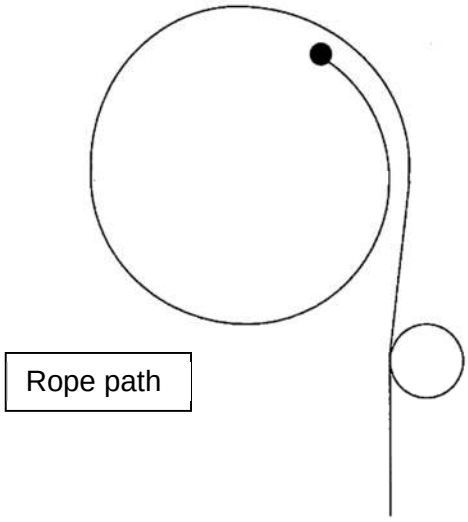




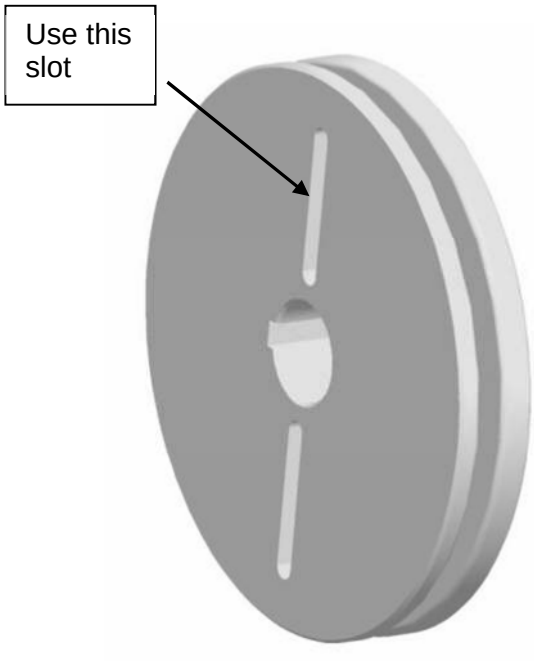
40. Door with pre-running photocell and breakaway: If the door is opened too high, the cable chain in the slider gets damaged.



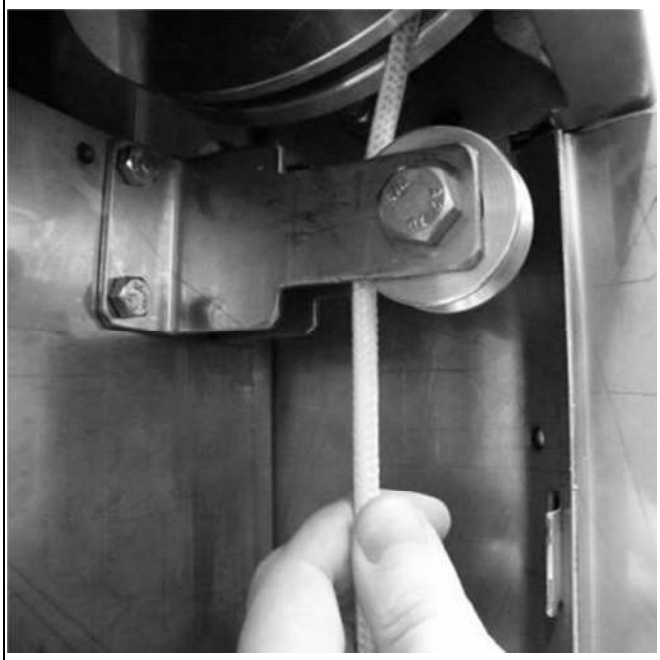
41. Door with counterweight: Feed the 7-mm rope from the pulley groove through the slot on the side of the pulley. Tie a knot in the rope. Pulley with 2 slots: **see Fig. 43.**



42. Door with counterweight: Attach the rope with minimum 1,1 but not more than 2 dead turns. As more dead turns as the door has, as longer is the way of the counterweight.



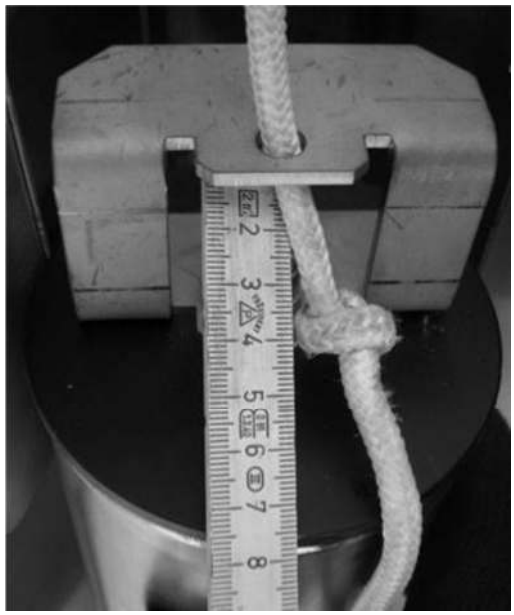
43. Door with counterweight: Some doors have pulleys with 2 slots to guide the rope through. Use the slot that results to approx. 1,1 to 1,6 dead turns.



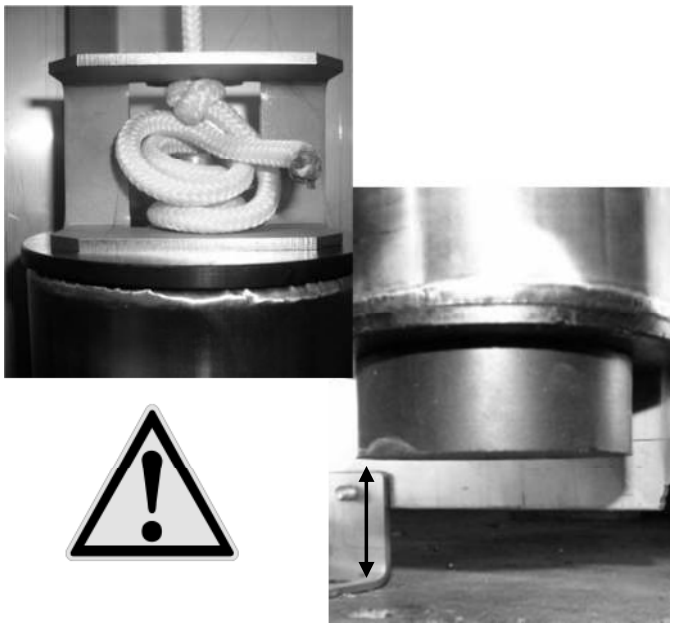
44. Door with counterweight: Feed the attached rope behind the pulley and downward.



45. Door with counterweight: Place the counterweight in the side frame on the wood block (H approx. 180mm is delivered with door). Hold the weight firmly and prevent it from tipping over.



46. Door with counterweight: Guide the 7mm rope through the hole of the bracket for weight. Pull the rope downwards and make a simple knot in the rope. Distance between the knot and the hole of the bracket: 40mm.



47. Door with counterweight: Lift the counterweight slightly and remove the wood block. The knot holds the counterweight in a height of approx. 50mm (bottom edge rubber bumper). Do not cut the rope.

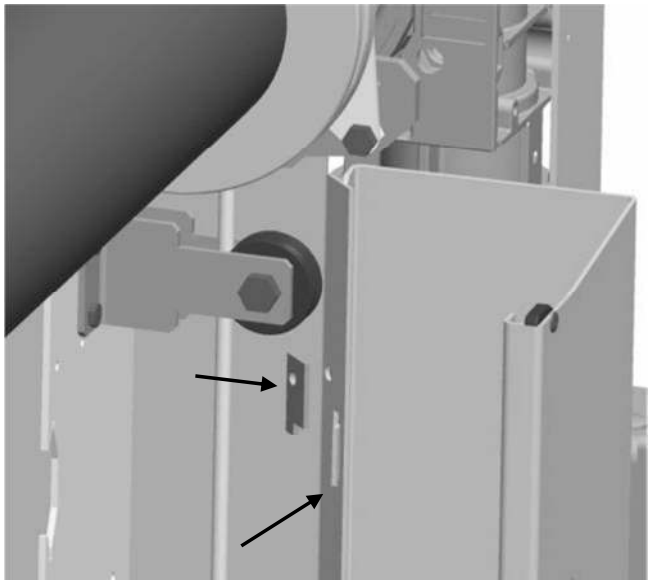


Caution: Make sure that the weight is not fitted too high. If the weight is fitted too high, it may collide with the pulley when the door reaches the end of its closing travel.

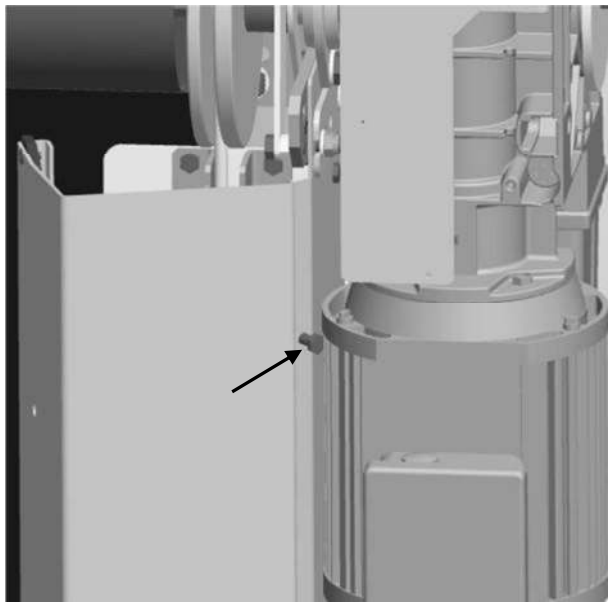
Do only open or close the door if the side frame cover is closed.

Caution: Make sure that the weight is not fitted too low. If the weight is fitted too low, the plastic panel of the weight may get stuck at the bottom of the side frame at the start of closing travel.

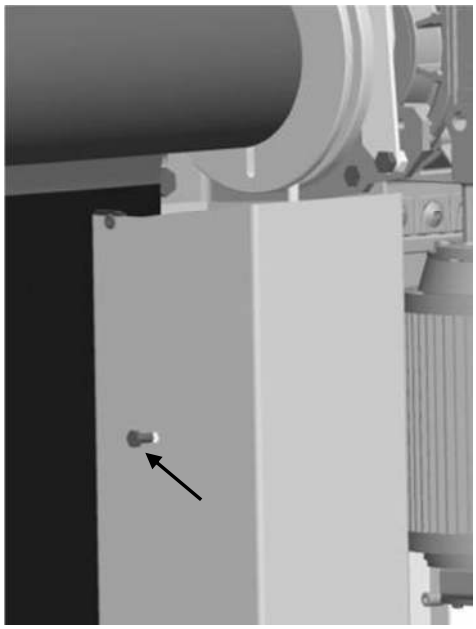
48. Door with counterweight



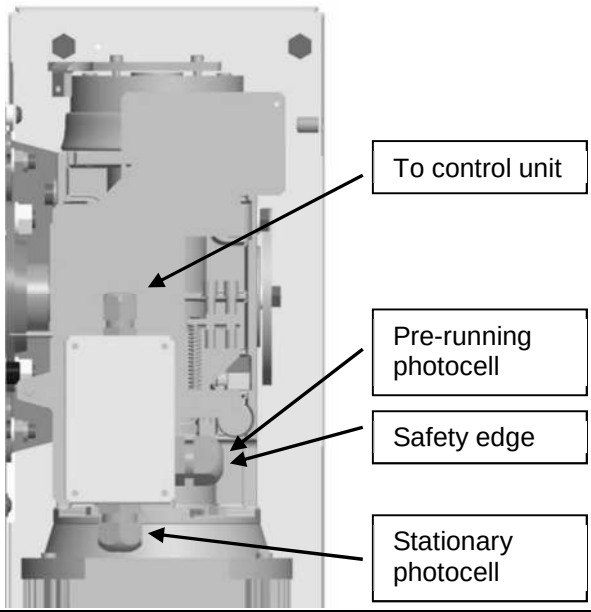
49. Insert the side frame cover into the side frame profile at an angle of approximately 45° with the tab of the cover profile at the bottom. To prevent the cover profile from falling out, refit the retainer which was removed in **Fig. 4**.



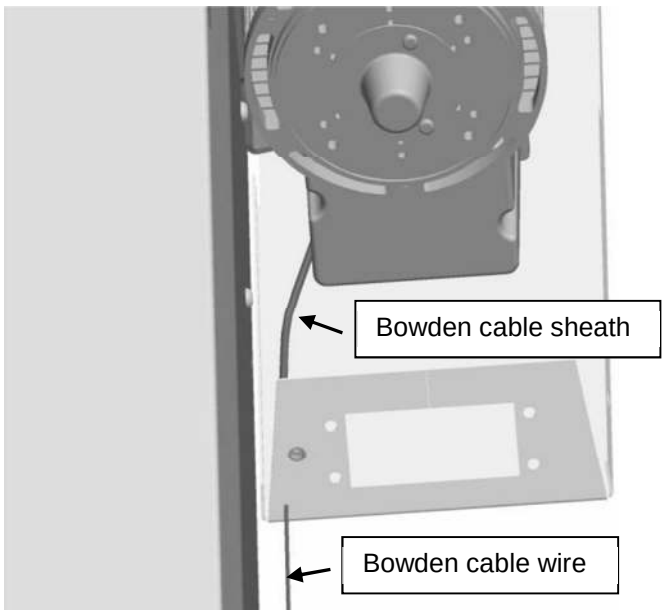
50. Secure the retainer with an M6x12 hex head screw and a cap nut M6.



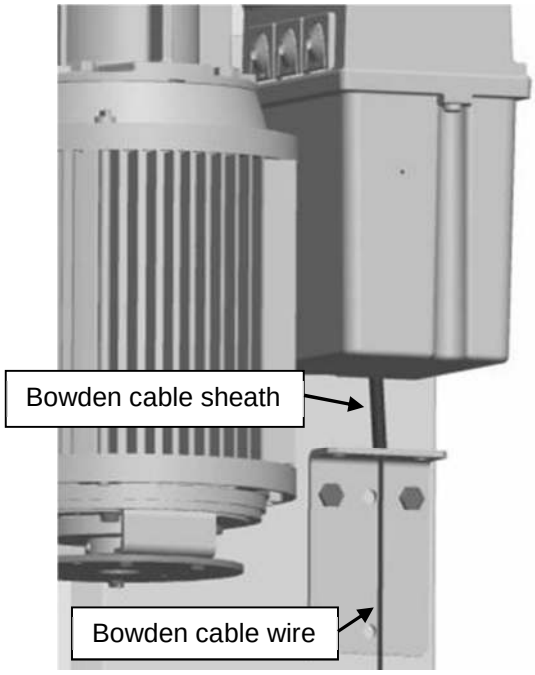
51. Side frame without easy-opening option: Fix with M6x12 hex head screws (illustrated). **Side frame with easy-opening option:** Lock the cover by pulling out and actuating the lever. **Secure by screwing an M6x12 hex head screw (see Fig. 6).**



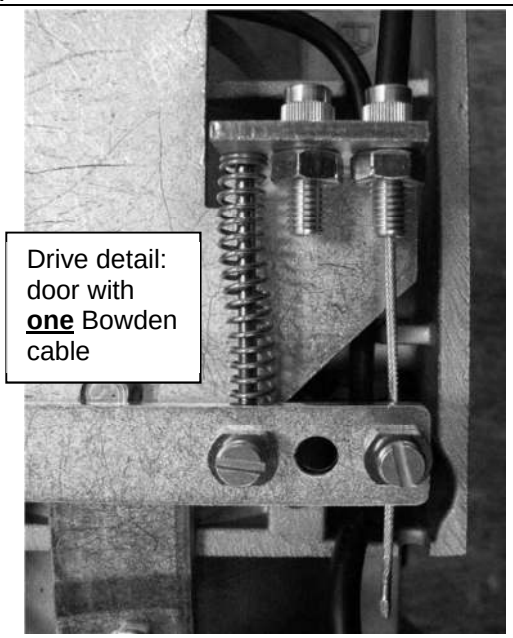
52. Using two cylinder-head screws M4x8, fasten the terminal box to the drive unit bracket. Wire according to the wiring diagram. The cable gland on the side must face outward.



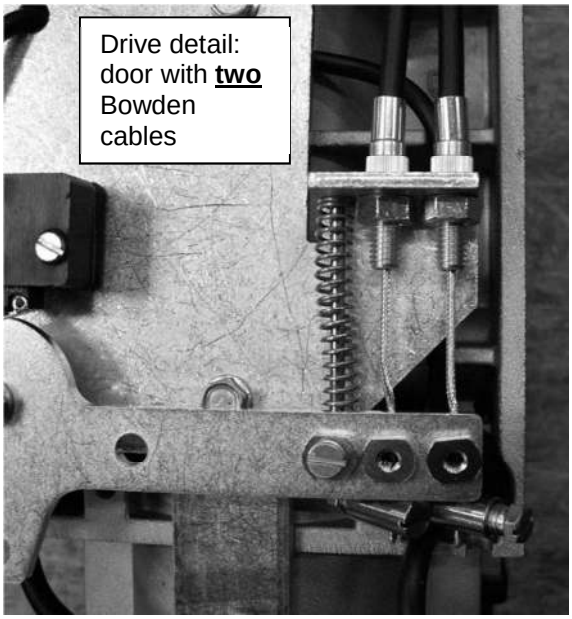
53. Door with motor cover and counterweight: Feed the Bowden cable wire through the motor cover bracket from below. Fit the Bowden cable sheath over the inner Bowden cable wire from above.



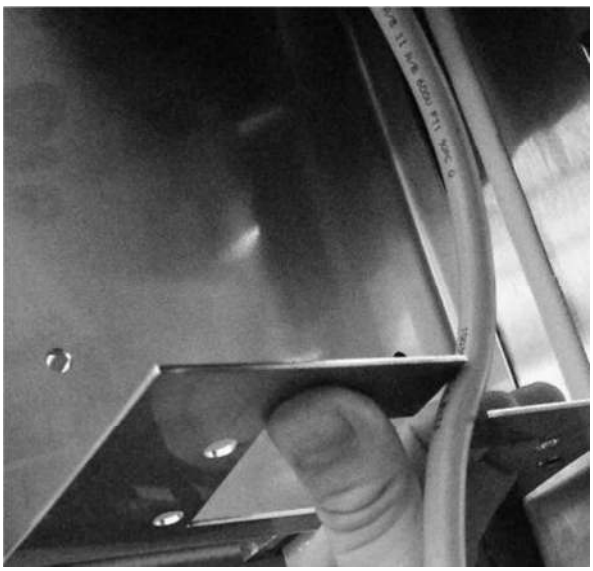
54. Door without motor cover and with counterweight: Feed the Bowden cable wire through the bracket on the side frame from below. Fit the Bowden cable sheath over the inner Bowden cable from wire above.



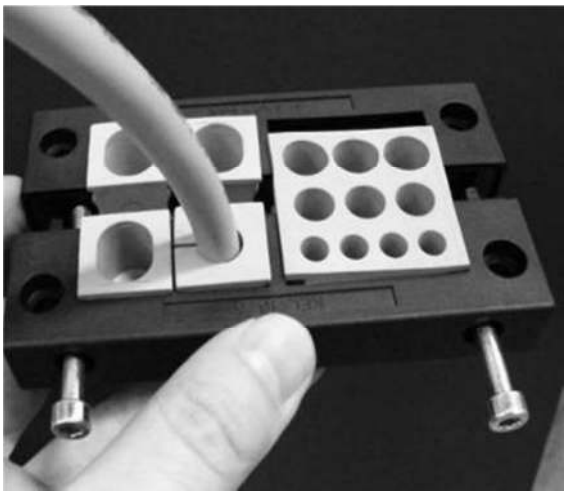
55. Door with counterweight: Feed the Bowden cable wire through the bore of the hollow screw. Feed the Bowden cable sheath from the side frame to the hollow screw. Clamp the Bowden cable wire to the lever with the screw. **Check that it works correctly.**



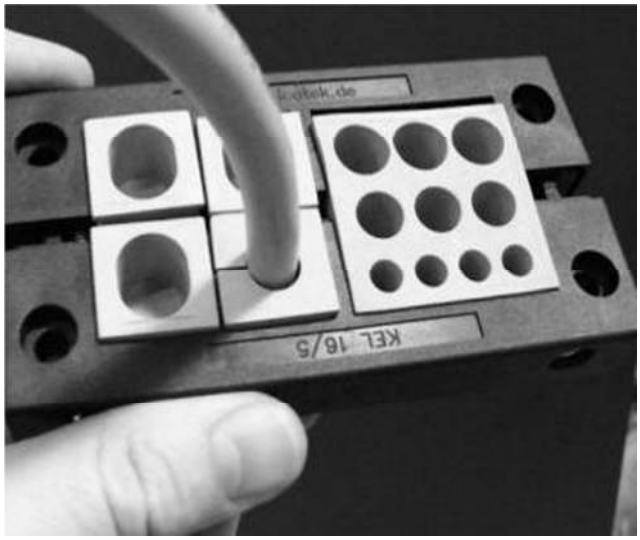
56. Door with counterweight: Feed the Bowden cable wires through the bores of the hollow screws. Feed the Bowden cable sheath from the side frame to the hollow screw. Do not clamp the Bowden cable wires to the lever. Using screws, clamp the Bowden cable wire below the hollow screw. **Check that it works correctly.**



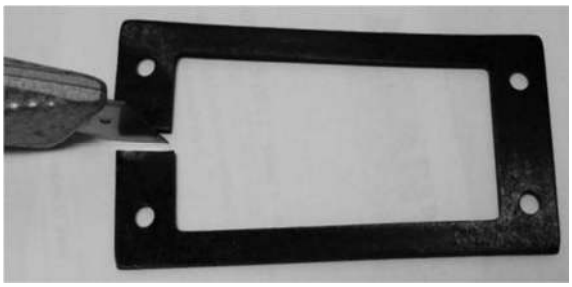
57. Door with motor cover: Feed the wires or cables from the terminal box to the control unit or the control buttons through the motor cover bracket.



58. Door with motor cover: Loosen the cable feedthrough. Slide suitable rubber inserts over the wire(s) or cable(s).



59. Door with motor cover: Push the cable through the cable feedthrough as far as necessary. Tighten the cable feedthrough**.**





60. Door with motor cover: Top picture: Cut an opening in the gasket on one of the short sides.

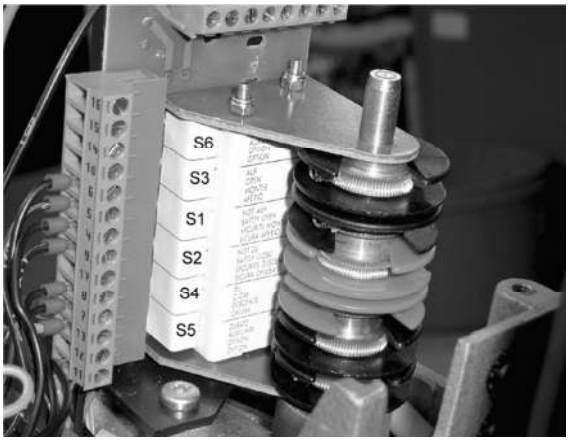
Bottom photo: Glue the gasket to the motor cover wall bracket.



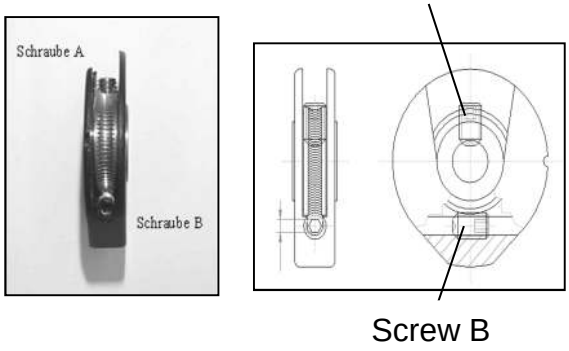


61. Door with motor cover: Feed the M6x25 hex head screws through the motor cover wall bracket from the below and secure them with washers and cap nuts fitted from above.

62. In order to adjust the stop position(s), you must remove the plastic cover from the drive unit.



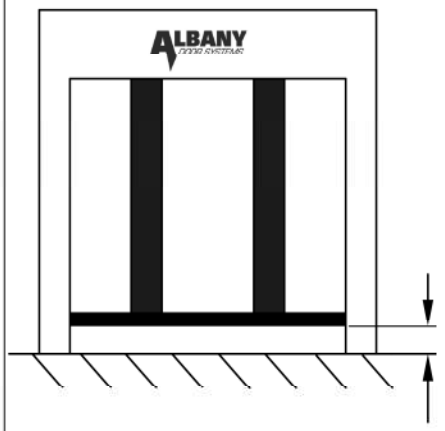
S1 Emerg. UP **S4** DOWN
S2 Emerg. DOWN
S5 CUTOFF STAT. PHOTOCELL
S3 UP **S6** ADDITIONAL



63. Connect the electrical cable(s) of the drive unit. Adjust the cam-operated limit switches according to the stop position. For top position see **Fig. 39.**
S4 and S5 are needed only with ACS 50. Caution: Incorrect settings can lead to damage!

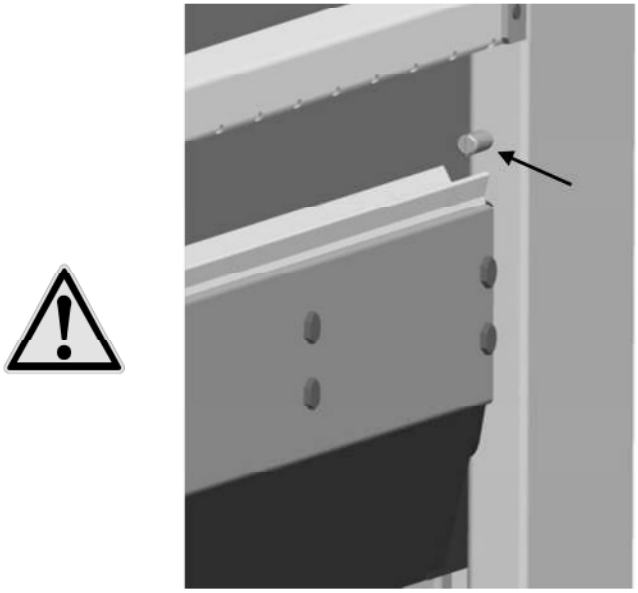
64. Once this has been roughly adjusted, tighten **screw A** and do not loosen again unless absolutely necessary (paint over to secure).

Use **screw B** for fine adjustment of the stop positions.

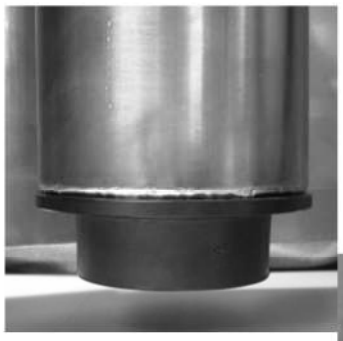


With ACS 50: Set bottom position with (S4)!

65. ACS50: Adjust the cam limit switch **S4** so that it is activated at an opening height of 30 – 50 mm. Doors with **MCC**: the lower end position and the shut-off point of the photocell are set via control unit. Mount the plastic covering cap at the drive unit.



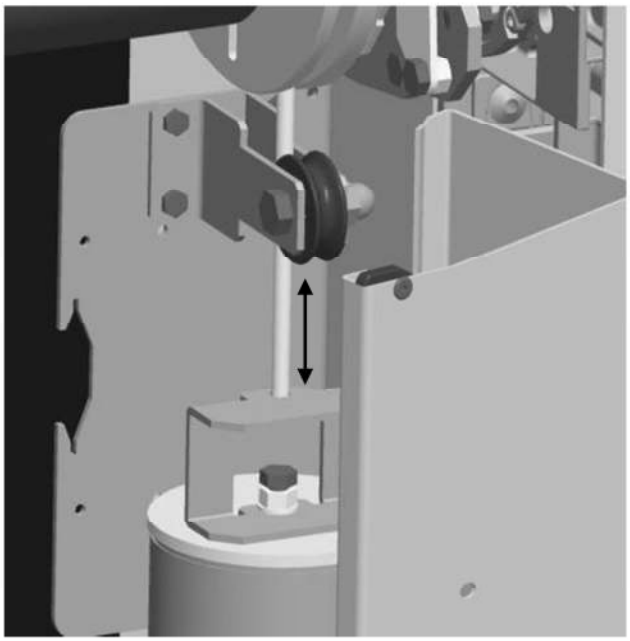
66. Door with pre-running photocell, with breakaway: Set the top position such high that the distance between bottom profile and stopper is approx. 5-10mm. **Door with pre-running photocell without breakaway:** Set the top position such high that the distance between bottom profile and stopper is approx. 20mm.



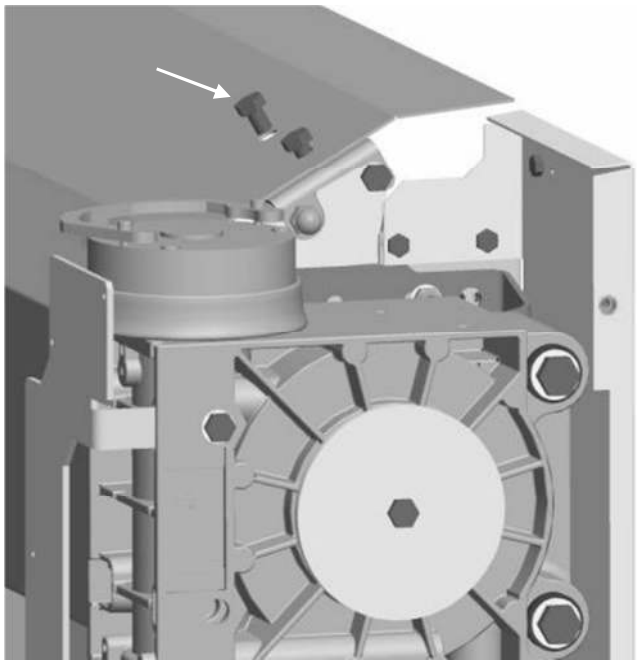


Do not open or close the door if the side frame cover is open

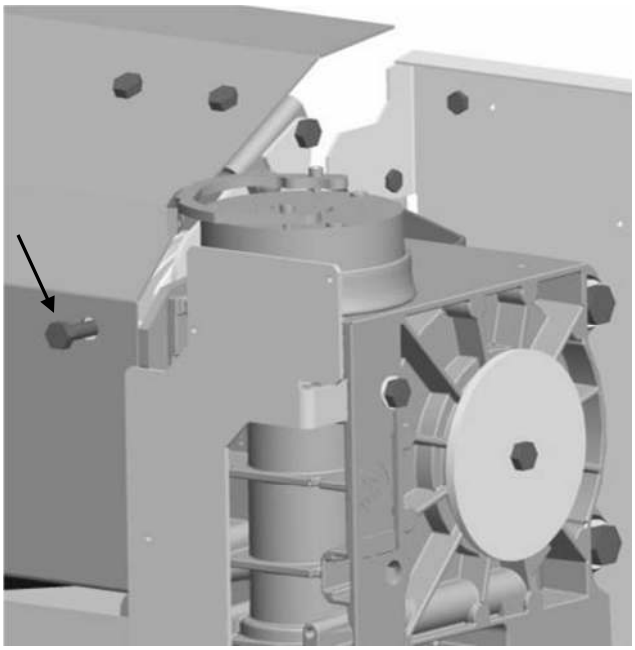
67. Door with counterweight: Run the door for approx. 10 cycles and check the height of the counterweight at open door. The counterweight (bottom edge rubber bumper) should be positioned at a height of 20mm. Maybe a correction of the position of the knot is necessary (**see Fig. 45 and 46**). Cut the rope.



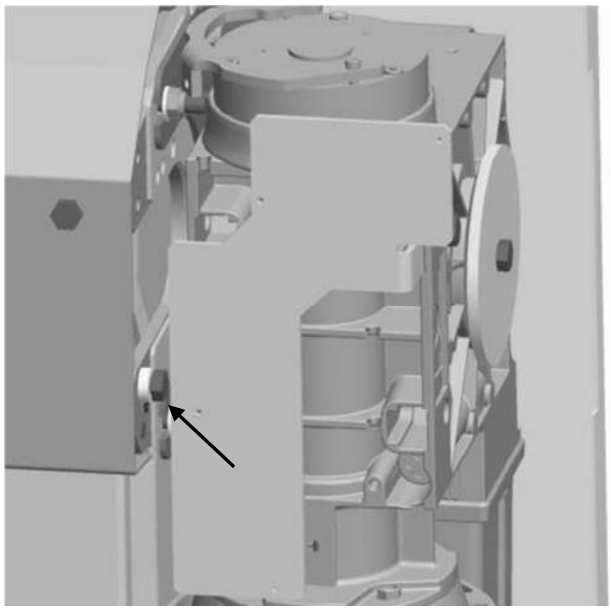
68. Door with counterweight: Increase the self opening. Increasing the self opening is possible by adding one extra dead turn. This is only possible if the space between the pulley and the bracket for counterweight is minimum 1/10 of the DH (e.g. 250mm at DH=2500).



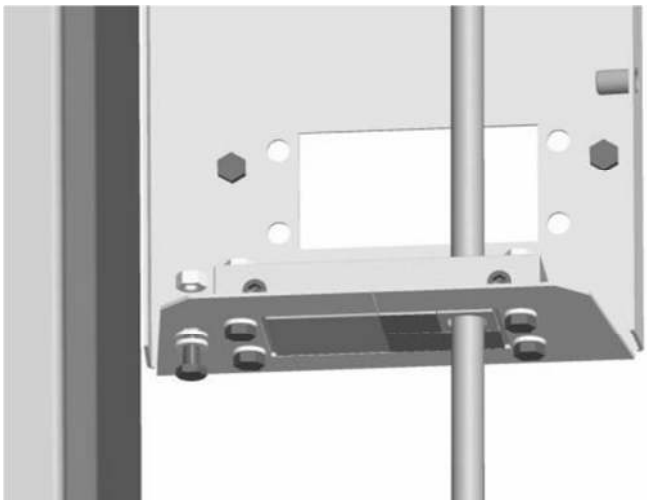
69. Door with top roll cover: Fix the top roll cover to the side frame members on each side with two M8x16 hex head screws at each end. Use a cap nut for each screw.



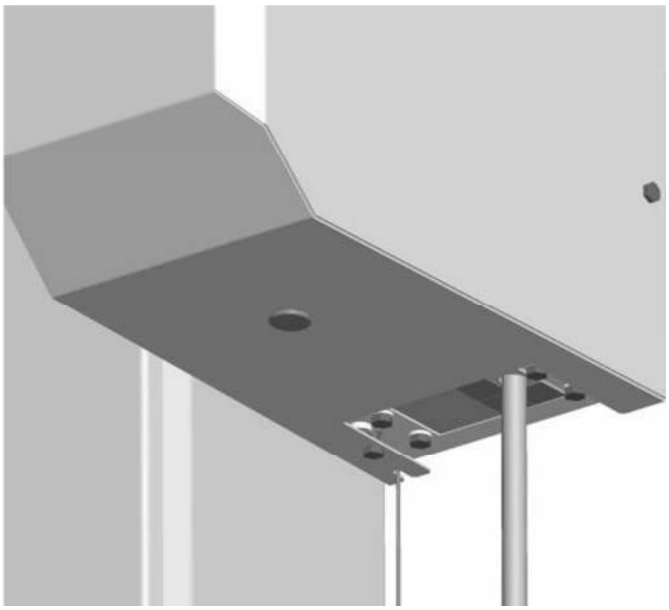
70. Door with top roll cover: Fit the retaining screws (M8x20 hex head screws with thin shaft).



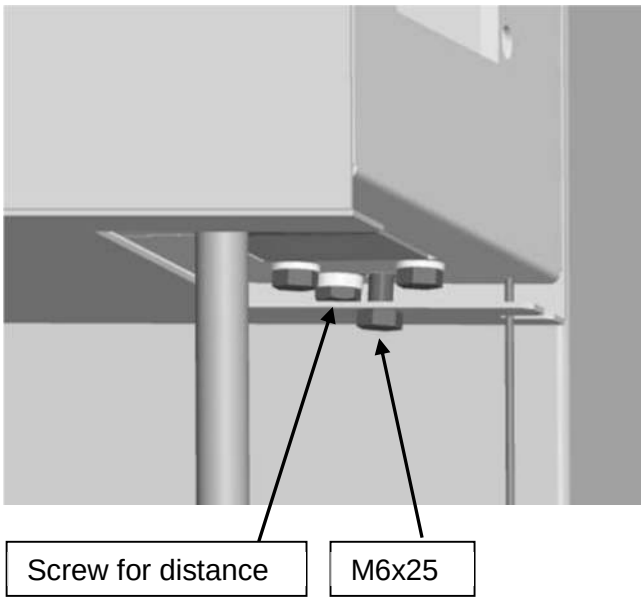
71. Connect equipotential bonding. Use a hex head screw M8x12 and three washers. Use a non-corroding cable eye.



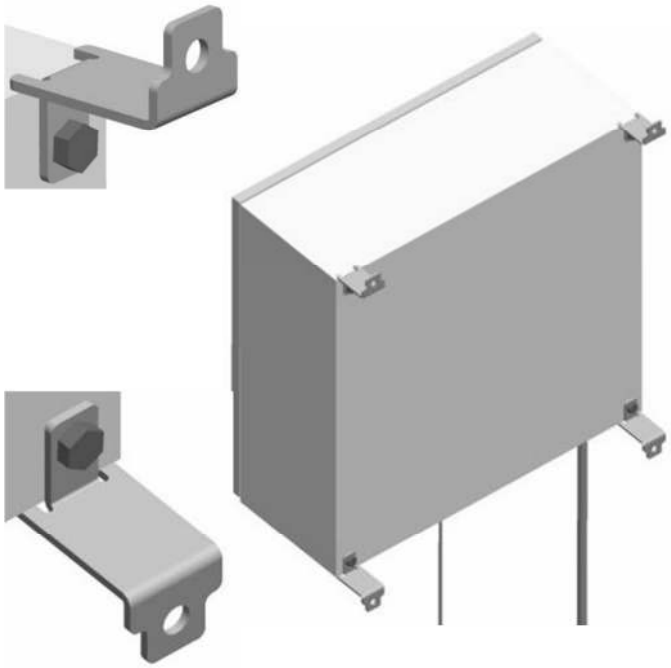
72. Door with motor cover: Insert a hex head screw M6x12 with 2 washers from below into the wall plate motor cover. Affix the screw with a cap nut.



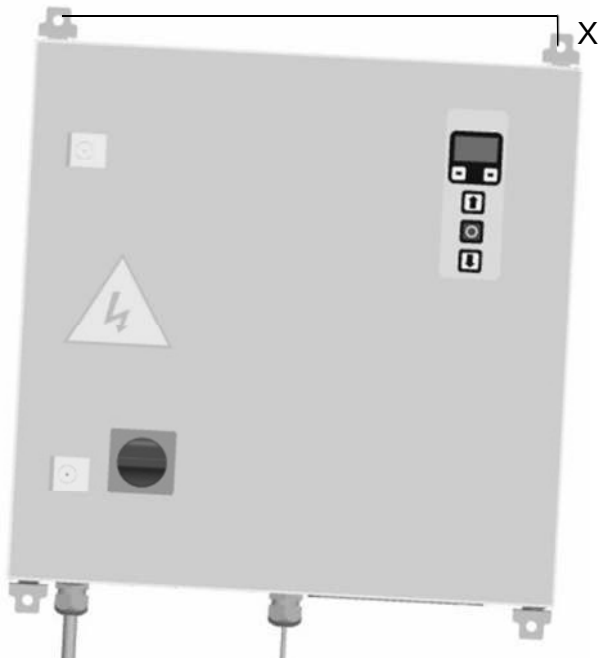
73. Door with motor cover: Slide the motor cover over the drive unit and the motor cover wall bracket. Secure the motor cover with three hex head screws (2x M6x12 from the side and 1x M6x25 from below).



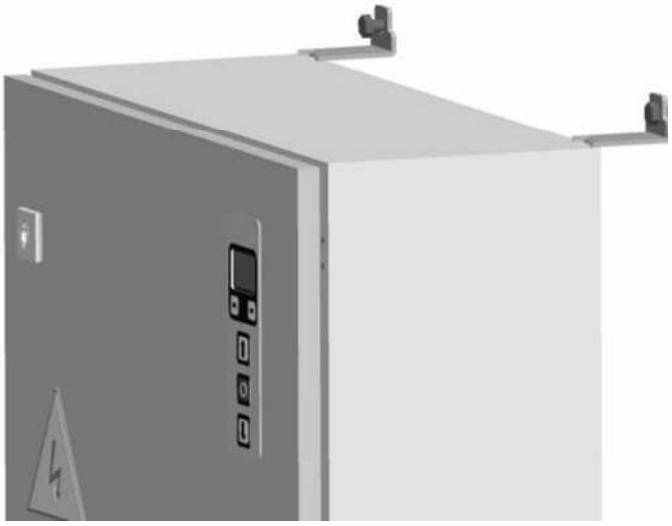
74. Door with motor cover: At the bottom of the motor cover a gap of approximately 7 mm must be between the motor cover bracket and the motor cover. Do not overtighten the M6x25 hex head screw.

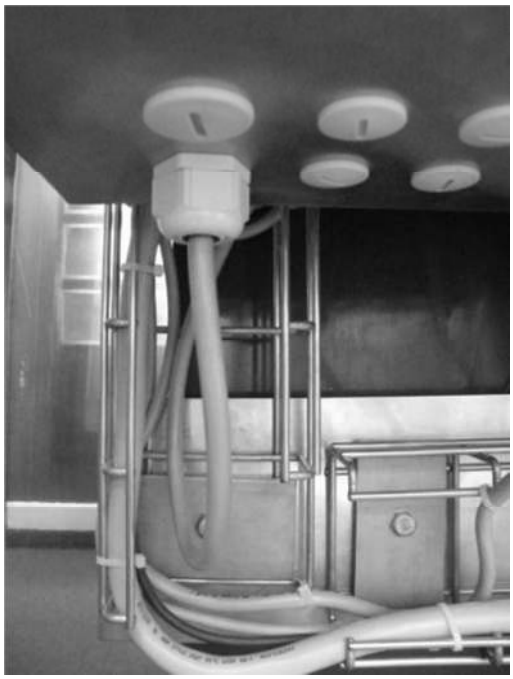


75. Door with control box: The 4 wall holders (installation bag) must be mounted to the control box. Use a hex head screw M8x12 and a nut for every wall holder.



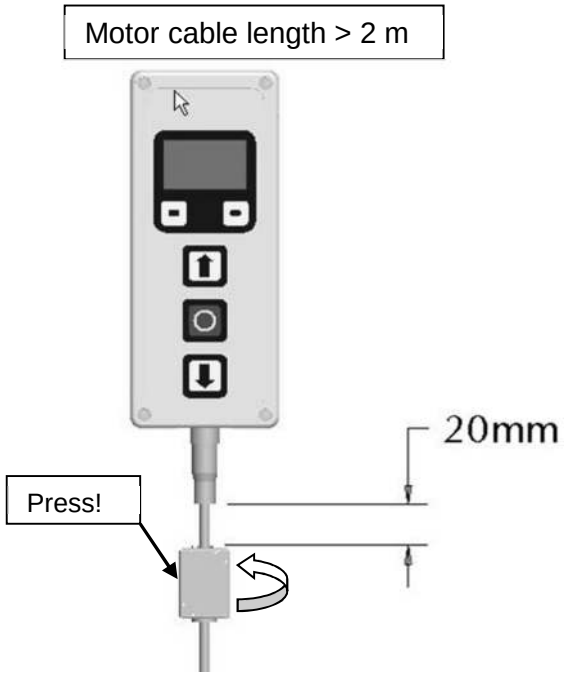
76. Door with control box: Fix the control box with a bevel drain of 3°. Control box 500x500: drill the holes displaced by 25mm (value X). Control box 380x380: drill the holes displaced by 18mm (value X).





77. Door with control box: fasten the control box with 4 screws to the wall. (These screws are not included with the door.) The cable fittings must be mounted on the bottom side of the control box.

78. Use open cable channels to pass the cables. Open cable channels can be delivered with the door (optional).



The assembly is finished after test of safety devices and test run.

The dismantling is made in opposite direction to installation.

The door packaging can be returned to the manufacturer.

Defective door components should be disposed at site according environmental regulations.

Defective electronic components must be disposed as special waste.

79. Door with MCC